

# **PMSCS 694**

## **Information Security**

### **Crypto Basics**

### **Lecture 3**

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# Crypto

- ❑ **Cryptology** — The art and science of making and breaking “secret codes”
- ❑ **Cryptography** — making “secret codes”
- ❑ **Cryptanalysis** — breaking “secret codes”
- ❑ **Crypto** — all of the above (and more)

# How to Speak Crypto

- ❑ A *cipher* or *cryptosystem* is used to *encrypt* the *plaintext*
- ❑ The result of encryption is *ciphertext*
- ❑ We *decrypt* ciphertext to recover plaintext
- ❑ A *key* is used to configure a cryptosystem
- ❑ A *symmetric key* cryptosystem uses the same key to encrypt as to decrypt
- ❑ A *public key* cryptosystem uses a *public key* to encrypt and a *private key* to decrypt

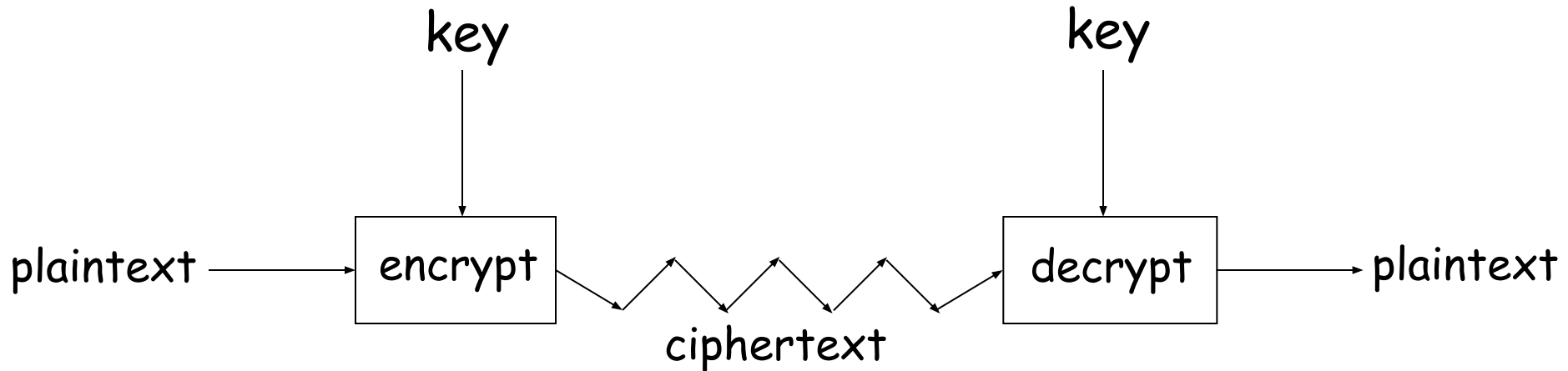
# Crypto

- ❑ Basic assumptions
  - The system is completely known to the attacker
  - Only the key is secret
  - That is, crypto algorithms are not secret
- ❑ “A cryptographic system should remain secure even if everything about the system, except the key, is public knowledge.”
- ❑ This is known as **Kerckhoffs' Principle**.
- ❑ Why do we make such an assumption?
  - Experience has shown that secret algorithms tend to be weak when exposed
  - Secret algorithms never remain secret
  - Better to find weaknesses beforehand

# Broader Interpretation of Kerckhoffs' Principle

- ❑ Generalized Definition: "A secure system should not rely on the secrecy of its design, but rather on the secrecy of small, easily changeable elements—such as passwords, keys, or credentials."
- ❑ This broader interpretation is often applied in software engineering, network design, and security system architecture, where the focus is on:
  - Open design: Encouraging peer review and testing
  - Rapid key rotation: Instead of redesigning entire systems
  - Defense in depth: Assuming attackers may understand the system layout
- ❑ Example: OpenSSL is a public, open-source cryptographic library. Despite its code being freely accessible, it remains secure because its strength lies in key secrecy, not in hidden implementation.

# Crypto as Black Box



*A generic view of symmetric key crypto*

# Simple Substitution

❑ Plaintext: **fourscoreandsevenyearsago**

❑ Key:

|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plaintext  | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
| Ciphertext | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |

❑ Ciphertext:

**IRXUVFRUHDQGVHYHQBHDUVDJR**

❑ Shift by 3 is "Caesar's cipher"

# Caesar's Cipher Decryption

- Suppose we know a Caesar's cipher is being used:

|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plaintext  | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
| Ciphertext | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |

- Given ciphertext:

VSRQJHEREVTXDUHSDQWV

- Plaintext: spongebobsquarepants



# Caesar's Cipher

- Shift by  $n$  for some  $n \in \{0,1,2,\dots,25\}$
- Then key is  $n$
- Example: key  $n = 7$

Plaintext

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
| H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |

Ciphertext

# Cryptanalysis I: Try Them All

- ❑ We know Caesar's cipher and shift by  $n$  used
  - But the specific key is unknown
- ❑ Given ciphertext: **CSYEVIXIVQMREXIH**
- ❑ How to determine the key?
- ❑ Only 26 possible keys —try them all!
- ❑ **Exhaustive key search**
- ❑ Solution: key is  $n = 4$

# Simple Substitution: General Case

- ❑ In general, simple substitution key can be any **permutation** of letters
  - Not necessarily a Caesar's cipher (shift)
- ❑ For example

|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plaintext  | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
| Ciphertext | J | I | C | A | X | S | E | Y | V | D | K | W | B | Q | T | Z | R | H | F | M | P | N | U | L | G | O |

- ❑ In general,  $26! > 2^{88}$  possible keys

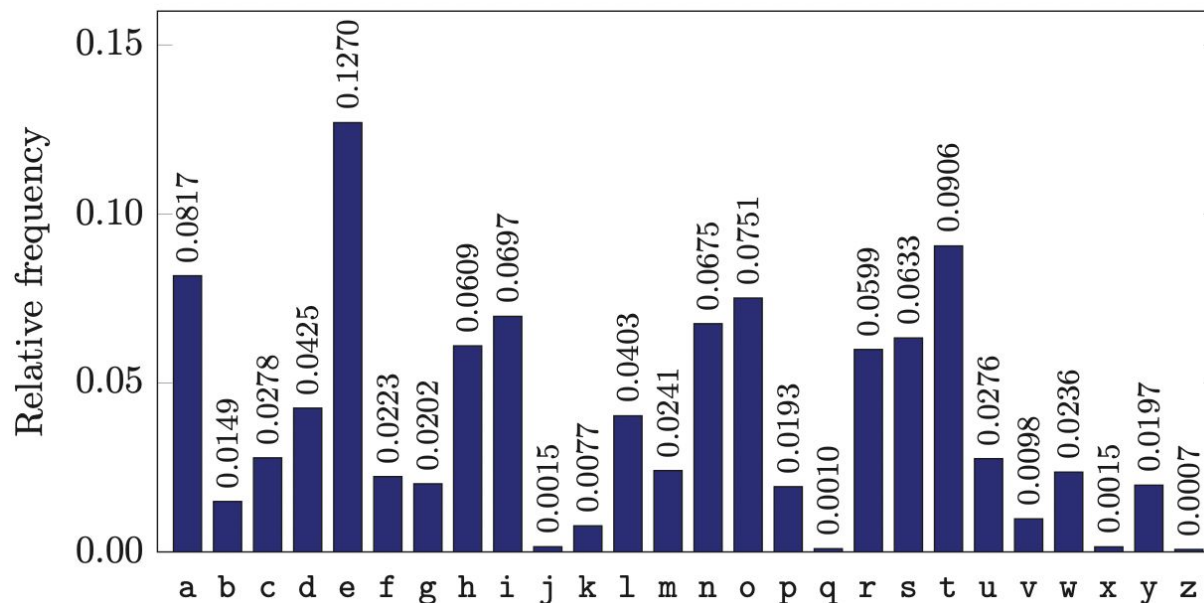
# Cryptanalysis II: Be Clever

- We know that a simple substitution used
- But *not* necessarily a Caesar's cipher (shift)
- Find the key given the ciphertext:

PBFPVYFBQXZTYFPBFEQJHDXXQVAPTPQJKTOYQWIPBVWLXTOX  
BTFXQWAXBVCXQWAXFQJVVWLEQNTQZQGGQLFXQWAKVWLXQ  
WAEBIPBFXFQVXGTVJVWLBTPQWAEBFPBFHCVLXBQUFEVWLXGD  
PEQVPQGVPFBFTIXPFHXZHVFAGFOTHFEBQUFTDHzBQPOTHXTY  
FTODXQHFTDPTOGHFQPBQWAQJJTODXQHFOQPWTBDHHIXQV  
APBFZQHCFWPFHPBFIPBQWKFABVYYDZBOTHBPQPQJTQOTOGHF  
QAPBFEQJHDXXQVAVXEBQPEFZBVFOJIWFFACFCFHQWAUVWF  
LQHGFVAFXQHFUFHILTAVWAFFAWTEVOITDHFHFQAITIXPFH  
XAFQHEFZQWGFLVWPTOFFA

# Cryptanalysis II

- ❑ Cannot try all  $2^{88}$  simple substitution keys
- ❑ Can we be more clever?
- ❑ English letter frequency counts...



# Cryptanalysis II

## □ Ciphertext:

PBFPVYFBQXZTYFPBFEQJHDXXQVAPTPQJKTOYQWIPBVWLXTOXBTFXQ  
WAXBVCXQWAXFQJ VWLEQNTQZQGGQLFXQWAKVWLXQWAEBIPBFXFQ  
VXGTVJVWLBTPQWAEFBFBFHCVLXBQUFEVWLXGDPEQVPQGVPPBFTIXPFH  
XZHVFA GFOTHFEBQUFTD HZBQPOTHXTYFTODXQHFTDPTOGHFQPBQW  
AQJJTODXQHFOQPWTBDHHIXQVAPBFZQHCFWPFHPBFIPBQWK FABVYY  
DZBOTHBPBPQJTQOTOGHFQAPBFEQJHDXXQVAVXEBQPEFZBVFOJIWFF  
ACFCCFHQWAUVWFLQHGF XVAFXQH FUFHILTTAVWAFFAWTEVOITDHFH  
FQAITIXPFH XAFQHEFZQWGFLVWPTOFFA

## □ Analyze this message using statistics below

Ciphertext frequency counts:

| A  | B  | C | D  | E  | F  | G  | H  | I  | J | K | L  | M | N | O  | P  | Q  | R | S | T  | U | V  | W  | X  | Y | Z |
|----|----|---|----|----|----|----|----|----|---|---|----|---|---|----|----|----|---|---|----|---|----|----|----|---|---|
| 21 | 26 | 6 | 10 | 12 | 51 | 10 | 25 | 10 | 9 | 3 | 10 | 0 | 1 | 15 | 28 | 42 | 0 | 0 | 27 | 4 | 24 | 22 | 28 | 6 | 8 |

# Cryptanalysis: Terminology

- ❑ Cryptosystem is **secure** if best know attack is to try all keys
  - Exhaustive key search, that is
- ❑ Cryptosystem is **insecure** if **any** shortcut attack is known
- ❑ But then insecure cipher might be harder to break than a secure cipher!
  - What the ... ?

# Vigenere Cipher

- ❑ Simple substitution is *monoalphabetic*
- ❑ Vigenere cipher is simple example of a *polyalphabetic* substitution
  - Caesars ciphers, based on a keyword
- ❑ For example, keyword CAT indicates shift by 2, shift by 0, shift by 19
  - Then repeat as needed



# Vigenere Example

- Suppose that we want to encrypt attackatdawn
- Encryption:

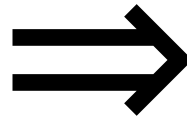
|             |              |
|-------------|--------------|
| keyword:    | CATCATCATCAT |
| plaintext:  | attackatdawn |
| ciphertext: | ctmccdctwcwg |
- Ciphertext is ctmccdctwcwg
- How to decrypt? How to attack?

# Double Transposition

- Plaintext: **attackxatxdawn**

|       | col 1 | col 2 | col 3 |
|-------|-------|-------|-------|
| row 1 | a     | t     | t     |
| row 2 | a     | c     | k     |
| row 3 | x     | a     | t     |
| row 4 | x     | d     | a     |
| row 5 | w     | n     | x     |

Permute rows  
and columns



|       | col 1 | col 3 | col 2 |
|-------|-------|-------|-------|
| row 3 | x     | t     | a     |
| row 5 | w     | x     | n     |
| row 1 | a     | t     | t     |
| row 4 | x     | a     | d     |
| row 2 | a     | k     | c     |

- Ciphertext: **xtawxnattxadakc**
- Key is matrix size and permutations:  
(3,5,1,4,2) and (1,3,2)

# One-Time Pad: Encryption

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Encryption:  $\text{Plaintext} \oplus \text{Key} = \text{Ciphertext}$

|             | h   | e   | i   | l   | h   | i   | t   | l   | e   | r   |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Plaintext:  | 001 | 000 | 010 | 100 | 001 | 010 | 111 | 100 | 000 | 101 |
| Key:        | 111 | 101 | 110 | 101 | 111 | 100 | 000 | 101 | 110 | 000 |
| Ciphertext: | 110 | 101 | 100 | 001 | 110 | 110 | 111 | 001 | 110 | 101 |
|             | s   | r   | l   | h   | s   | s   | t   | h   | s   | r   |

# One-Time Pad: Decryption

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

**Decryption:** Ciphertext  $\oplus$  Key = Plaintext

|             | s   | r   | l   | h   | s   | s   | t   | h   | s   | r   |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ciphertext: | 110 | 101 | 100 | 001 | 110 | 110 | 111 | 001 | 110 | 101 |
| Key:        | 111 | 101 | 110 | 101 | 111 | 100 | 000 | 101 | 110 | 000 |
| Plaintext:  | 001 | 000 | 010 | 100 | 001 | 010 | 111 | 100 | 000 | 101 |
|             | h   | e   | i   | l   | h   | i   | t   | l   | e   | r   |

# One-Time Pad

Double agent claims following "key" was used:

|              | s   | r   | l   | h   | s   | s   | t   | h   | s   | r   |
|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ciphertext:  | 110 | 101 | 100 | 001 | 110 | 110 | 111 | 001 | 110 | 101 |
| "key":       | 101 | 111 | 000 | 101 | 111 | 100 | 000 | 101 | 110 | 000 |
| "Plaintext": | 011 | 010 | 100 | 100 | 001 | 010 | 111 | 100 | 000 | 101 |
|              | k   | i   | l   | l   | h   | i   | t   | l   | e   | r   |

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

# One-Time Pad

Or, might claim the key is...

|              | s   | r   | l   | h   | s   | s   | t   | h   | s   | r   |
|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ciphertext:  | 110 | 101 | 100 | 001 | 110 | 110 | 111 | 001 | 110 | 101 |
| "key":       | 111 | 101 | 000 | 011 | 101 | 110 | 001 | 011 | 101 | 101 |
| "Plaintext": | 001 | 000 | 100 | 010 | 011 | 000 | 110 | 010 | 011 | 000 |
|              | h   | e   | l   | i   | k   | e   | s   | i   | k   | e   |

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

# One-Time Pad Summary

- ❑ **Provably** secure
  - Ciphertext gives **no** useful info about plaintext
  - All plaintexts are *equally likely*
- ❑ BUT, only when be used correctly
  - Pad must be random, used only once
  - Pad is known only to sender and receiver
- ❑ Note: pad (key) is same size as message
- ❑ So, why not distribute message itself, instead of the pad?

# Real-World One-Time Pad

- ❑ Project VENONA
  - Soviet spies encrypted messages from U.S. to Moscow in 30's, 40's, and 50's
  - Nuclear espionage, etc.
  - Thousands of messages
- ❑ Spy carried one-time pad into U.S.
- ❑ Spy used pad to encrypt secret messages
- ❑ Repeats within the "one-time" pads made cryptanalysis possible



# VENONA Decrypt (1944)

[C% Ruth] learned that her husband [v] was called up by the army but he was not sent to the front. He is a mechanical engineer and is now working at the ENORMOUS [ENORMOZ] [vi] plant in SANTA FE, New Mexico. [45 groups unrecoverable]

detain VOLOK [vii] who is working in a plant on ENORMOUS. He is a FELLOWCOUNTRYMAN [ZEMLYaK] [viii]. Yesterday he learned that they had dismissed him from his work. His active work in progressive organizations in the past was cause of his dismissal. In the FELLOWCOUNTRYMAN line LIBERAL is in touch with CHESTER [ix]. They meet once a month for the payment of dues. CHESTER is interested in whether we are satisfied with the collaboration and whether there are not any misunderstandings. He does not inquire about specific items of work [KONKRETNAYa RABOTA]. In as much as CHESTER knows about the role of LIBERAL's group we beg consent to ask C. through LIBERAL about leads from among people who are working on ENOURMOUS and in other technical fields.

- ❑ "Ruth" == Ruth Greenglass
- ❑ "Liberal" == Julius Rosenberg
- ❑ "Enormous" == the atomic bomb

# Codebook Cipher

- ❑ Literally, a book filled with “codewords”
- ❑ Zimmerman Telegram encrypted via codebook

|                |       |
|----------------|-------|
| Februar        | 13605 |
| fest           | 13732 |
| finanzielle    | 13850 |
| folgender      | 13918 |
| Frieden        | 17142 |
| Friedenschluss | 17149 |

: :

- ❑ Modern block ciphers are codebooks!
- ❑ More about this later...

# Codebook Cipher: Additive

- ❑ Codebooks also (usually) use **additive**
- ❑ Additive —book of “random” numbers
  - Encrypt message with codebook
  - Then choose position in additive book
  - Add additive sequence to get ciphertext
  - Send ciphertext and additive position (MI)
  - Recipient subtracts additives before decrypting
- ❑ Why use an additive sequence?

# Zimmermann Telegram

- Perhaps most famous codebook ciphertext ever
- A major factor in U.S. entry into World War I

CLASS OF SERVICE DESIRED  
☒ Post Day Message  
☐ Day Letter  
☐ Night Message  
☐ Night Letter

Persons should mark as it may use the above if available. OTHERWISE THE TELEGRAM WILL BE TRANSMITTED AS A FAST DAY MESSAGE.

**WESTERN UNION**  
**TELEGRAM**

NEWCOMB CARLTON, PRESIDENT

Send the following telegram, subject to the terms on back hereof, which are hereby agreed to

via Galveston

JAN 19 1917

GERMAN LEGATION  
MEXICO CITY

|       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 130   | 13042 | 13401 | 8501  | 115   | 3528  | 416   | 17214 | 8491  | 11310 |
| 18147 | 18222 | 21560 | 10247 | 11518 | 23677 | 13605 | 3494  | 14936 |       |
| 98092 | 5905  | 11311 | 10392 | 10371 | 0302  | 21290 | 5161  | 39695 |       |
| 23571 | 17504 | 11269 | 18276 | 18101 | 0317  | 0228  | 17694 | 4473  |       |
| 22284 | 22200 | 19452 | 21589 | 67893 | 5569  | 13918 | 8958  | 12137 |       |
| 1333  | 4725  | 4458  | 5905  | 17166 | 13851 | 4458  | 17149 | 14471 | 6708  |
| 13850 | 12224 | 6929  | 14991 | 7382  | 15857 | 67893 | 14218 | 36477 |       |
| 5870  | 17553 | 67893 | 5870  | 5454  | 16102 | 15217 | 22801 | 17138 |       |
| 21001 | 17388 | 7446  | 23638 | 18222 | 6719  | 14331 | 15021 | 23845 |       |
| 3156  | 23552 | 22096 | 21604 | 4797  | 9497  | 22464 | 20855 | 4377  |       |
| 23610 | 18140 | 22260 | 5905  | 13347 | 20420 | 39689 | 13732 | 20667 |       |
| 6929  | 5275  | 18507 | 52262 | 1340  | 22049 | 13339 | 11265 | 22295 |       |
| 10439 | 14814 | 4178  | 6992  | 8784  | 7632  | 7357  | 6926  | 52262 | 11267 |
| 21100 | 21272 | 9346  | 9559  | 22464 | 15874 | 18502 | 18500 | 15857 |       |
| 2188  | 5376  | 7381  | 98092 | 16127 | 13486 | 9350  | 9220  | 76036 | 14219 |
| 5144  | 2831  | 17920 | 11347 | 17142 | 11264 | 7667  | 7762  | 15099 | 9110  |
| 10482 | 97556 | 3569  | 3670  |       |       |       |       |       |       |

BEPNSTORFF.

Charge German Embassy.

# Zimmermann Telegram Decrypted

- ❑ British had recovered partial codebook
- ❑ Then able to fill in missing parts

RECEIVED  
October 1-8-58  
Washington, State Dept.  
By *Wm. A. Eckhoff, Assistant*  
Date *Oct. 27, 1918*

TELEGRAM RECEIVED.

FROM 2nd from London # 5747.

"We intend to begin on the first of February unrestricted submarine warfare. We shall endeavor in spite of this to keep the United States of America neutral. In the event of this not succeeding, we make Mexico a proposal of alliance on the following basis: make war together, make peace together, generous financial support and an understanding on our part that Mexico is to reconquer the lost territory in Texas, New Mexico, and Arizona. The settlement in detail is left to you. You will inform the President of the above most secretly as soon as the outbreak of war with the United States of America is certain and add the suggestion that he should, on his own initiative, ~~invite~~ <sup>invite</sup> Japan to immediate adherence and at the same time mediate between Japan and ourselves. Please call the President's attention to the fact that the ruthless employment of our submarines now offers the prospect of compelling England in a few months to make peace." Signed, ZIMMERMANN.

# Random Historical Items

- ❑ Crypto timeline
- ❑ Spartan Scytale — transposition cipher
- ❑ Caesar's cipher
- ❑ Poe's short story: *The Gold Bug*
- ❑ Election of 1876



# Election of 1876

- ❑ “Rutherfraud” Hayes vs “Swindling” Tilden
  - Popular vote was virtual tie
- ❑ Electoral college delegations for 4 states (including Florida) in dispute
- ❑ Commission gave all 4 states to Hayes
  - Voted on straight party lines
- ❑ Tilden accused Hayes of bribery
  - Was it true?

# Election of 1876

- ❑ Encrypted messages by Tilden supporters later emerged
- ❑ Cipher: Partial codebook, plus transposition
- ❑ Codebook substitution for important words

## ciphertext

Copenhagen

Greece

Rochester

Russia

**Warsaw**

:

## plaintext

Greenbacks

Hayes

votes

Tilden

**telegram**

:



# Election of 1876

- ❑ Apply codebook to original message
- ❑ Pad message to multiple of 5 words (total length, 10,15,20,25 or 30 words)
- ❑ For each length, a fixed permutation applied to resulting message
- ❑ Permutations found by comparing several messages of same length
- ❑ Note that the **same key** is applied to all messages of a given length

# Election of 1876

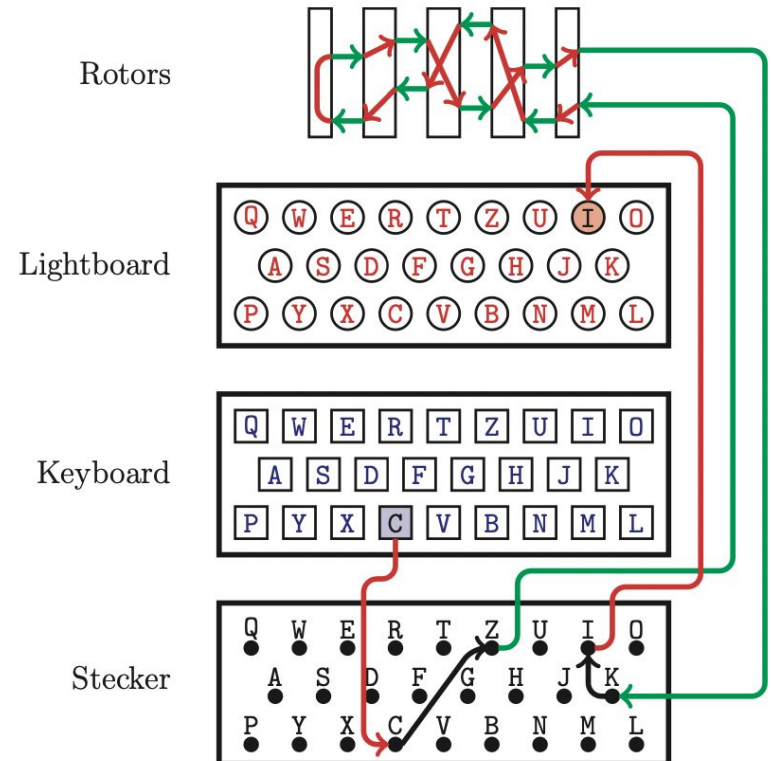
- ❑ Ciphertext: **Warsaw they read all unchanged last are idiots can't situation**
- ❑ Codebook: Warsaw == telegram
- ❑ Transposition: 9,3,6,1,10,5,2,7,4,8
- ❑ Plaintext: **Can't read last telegram. Situation unchanged. They are all idiots.**
- ❑ A weak cipher made worse by reuse of key
- ❑ Lesson? Don't overuse keys!

# Early 20th Century

- ❑ WWI —Zimmermann Telegram
- ❑ "Gentlemen do not read each other's mail"
  - Henry L. Stimson, Secretary of State, 1929
- ❑ WWII —golden age of cryptanalysis
  - Midway/Coral Sea
  - Japanese Purple (codename MAGIC)
  - German Enigma (codename ULTRA)

# Enigma Cipher Machine

- ❑ Most famous cipher of WWII
  - Electro-mechanical device
  - Very rugged, used in the field



# Post-WWII History

- ❑ Claude Shannon —father of the science of information theory
- ❑ Computer revolution —lots of data to protect
- ❑ Data Encryption Standard (DES), 70's
- ❑ Public Key cryptography, 70's
- ❑ CRYPTO conferences, 80's
- ❑ Advanced Encryption Standard (AES), 90's
- ❑ The crypto genie is out of the bottle...

# Claude Shannon

- ❑ Founded field of information theory
- ❑ His 1949 paper: *Comm. Thy. of Secrecy Systems*
- ❑ Fundamental concepts
  - **Confusion** —obscure relationship between plaintext and ciphertext
  - **Diffusion** —spread plaintext statistics through the ciphertext
- ❑ Proved one-time pad is secure
- ❑ One-time pad is confusion-only, while double transposition is diffusion-only

# Taxonomy of Cryptography

## □ Symmetric Key

- Same key for encryption and decryption
- Modern types: Stream ciphers, Block ciphers

## □ Public Key (or “asymmetric” crypto)

- Two keys, one for encryption (public), and one for decryption (private)
- And digital signatures —nothing comparable in symmetric key crypto

## □ Hash algorithms

- Can be viewed as “one way” crypto

# Taxonomy of Cryptanalysis

- From perspective of info available to Trudy...
  - Ciphertext only — Trudy's worst-case scenario
  - Known plaintext
  - Chosen plaintext
    - "Lunchtime attack"
    - Some protocols will encrypt chosen data
  - Adaptively chosen plaintext
  - Related key
  - Forward search (public key crypto)
  - And others...



# Sample Questions

- ❑ What is Kerckhoffs' Principle? Why is it important in cryptographic system design?
- ❑ Describe the basic process of encryption and decryption in a cryptosystem.
- ❑ Explain the difference between symmetric and public key cryptosystems.
- ❑ What is a Caesar cipher? Describe its encryption and decryption processes.
- ❑ Discuss the limitations of the Caesar cipher and how it can be attacked.
- ❑ A plaintext message "HELLO" is encrypted using a Caesar cipher with a shift of 3. Write the ciphertext.
- ❑ A Caesar cipher with a shift of  $n=4$  produces the ciphertext "CSYEVIXIVQMREXIH". Decrypt this ciphertext to recover the plaintext

# Sample Questions

- ❑ What is the total number of possible keys in a Caesar cipher? Show the calculation.
- ❑ Explain the difference between symmetric and public key cryptosystems.
- ❑ Encrypt the message "welcome to cse ju" using a double transposition cipher with 4 rows and 4 columns, using the row permutation  $(1, 2, 3, 4) \rightarrow (2, 4, 1, 3)$  and the column permutation  $(1, 2, 3, 4) \rightarrow (3, 1, 2, 4)$ .
- ❑ Suppose you are given a one-time pad = 111 101 110 101 111 100 and alphabet encodings,
- ❑ Now determine the ciphertext for the following plaintexts
  - THRILL
  - TILLER

| ALPHABET | E   | H   | I   | K   | L   | R   | S   | T   |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|
| BINARY   | 000 | 001 | 010 | 011 | 100 | 101 | 110 | 111 |